Wentao Zhang
Research Assistant, Power Electronics and Machines Centre
Faculty of Engineering, University of Nottingham
wentao.zhang@nottingham.ac.uk · (+44) 7402 114 543

14 May 2026

Dr Judith Apsley and Members of the Selection Committee
Department of Electrical and Electronic Engineering
School of Engineering, The University of Manchester
Oxford Road, Manchester M13 9PL, United Kingdom

Dear Dr Apsley and Members of the Selection Committee,

Application for Lecturer / Senior Lecturer in Electrical Machines, SAE-031274

I am writing to apply for the **Lecturer / Senior Lecturer in Electrical Machines** post (**SAE-031274**) in the Department of Electrical and Electronic Engineering at the **University of Manchester**. I am currently a Research Assistant at the **Power Electronics and Machines Centre (PEMC), University of Nottingham**, having completed my **PhD in Electrical Engineering at Southeast University (SEU), Nanjing**, in September 2025 with the dissertation *Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles*. My research has focused on the electromagnetic design, thermal management, multi-physics optimisation and control of high-power-density electrical machines, and I would be glad to contribute to the Department's research and teaching in this area.

My research record to date includes **22 journal papers, of which 17 are in IEEE Transactions** (across *IEEE Trans. Industrial Electronics*, *Transportation Electrification*, *Industry Applications*, and *Energy Conversion*), 18 peer-reviewed conference papers at IEEE IEMDC, IEEE ECCE, ICEM and ICEMS, and 9 patents. The work covers novel *n*-layer flat-wire and hairpin winding design, electromagnetic analysis of flux-modulation and hybrid-excitation machines, high-fidelity thermal-network and oil-cooling modelling, multi-objective multi-physics optimisation, and converter-machine interaction under PWM excitation. This work has been recognised by the **2025 IEEE TEC Electric Vehicle Design Competition (1st Prize)**, the **2025 IEEE Trans. Transportation Electrification Prize Paper Award (2nd Prize)**, the **1st Prize in the IEEE IAS CMD Thesis Contest (non-PhD category)**, the **CIEEC Best Paper Award (2023)**, and the **Jiangsu Province Outstanding Master Degree Thesis Award (2022)**. I also serve as Co-Organiser and Co-Chair of an ICEM 2026 Special Session on *Next-Generation Electrified Transport*.

I have also gained substantive industrial-development experience through a collaboration with **Inovance**, a leading EV traction-machine supplier, on two oil-cooled PM traction machines, an **18 000 rpm 113 kW A0-class machine** (138 Nm, 7.78 kW/kg, 95.3% peak efficiency) and a **20 000 rpm 270 kW A-class machine** (380 Nm, 7.87 kW/kg, 97.4% peak efficiency). Each was taken through a full industrial cycle of multi-physics design (electromagnetic FEA, lumped-parameter thermal-network modelling and mechanical-stress evaluation), engineering drawings and tolerance specification for high-reliability manufacture, and AVL dynamometer characterisation. Alongside this, I have developed an **in-house FEA code package** covering **electromagnetic, structural-modal and mechanical-stress analyses**, now being applied to magnet-free synchronous-machine work at PEMC. This combined academic and industrial background should support applications for **EPSRC**, **Innovate UK** and **Aerospace Technology Institute (ATI)** funding, as well as direct industrial co-funded R&D with EV and aerospace machine manufacturers.

I would be happy to contribute to teaching across the **Electrical Machines and Drives curriculum** at undergraduate and MSc level, including **electromagnetic machine theory, finite-element principles, FEA-based machine design, thermal management, and machine control**, as well as converter-machine interaction. I would also be glad to develop new teaching content on contemporary traction-machine technology such as **hairpin winding and advanced cooling**, which is of direct relevance to students pursuing careers in EV and aerospace electrification. During my doctoral studies I served as **teaching assistant** on the SEU course *Integrated Practice of Electrical Machine Systems*, preparing tutorial material, leading Q&A, reviewing project proposals, and contributing mark recommendations. My toolchain spans MATLAB/Simulink, ANSYS

Maxwell and Mechanical, Python, C/C++ for TI C2000 and STM32, and FPGA, which should be useful for project-based teaching and capstone supervision.

Although cryogenic machine design has not been my primary research focus, my thermal-modelling background, including **high-fidelity lumped-parameter networks and coupled fluid-thermal analyses**, provides a foundation that I would be glad to extend alongside Manchester's **superconducting and HTS-machine activities**. For laboratory health and safety, I have worked in **high-voltage electrical-machine test facilities at SEU and PEMC under formal H&S frameworks**, including risk assessment, laboratory induction and safe working procedures for rotating machinery and power-electronics test rigs, and I appreciate the responsibility of managing such environments for student use. My research also has a substantial power-electronics component, addressing **PWM-excitation effects on machine losses, field-winding voltage modulation, and induced-voltage pulsation**, and I have designed and built a **three-phase full-bridge converter with associated control** for flux-modulation machine testing, which provides a basis for contributing to power-electronics and AC power-circuit analysis teaching.

My CV and detailed responses to the person specification are attached. I would be glad to discuss my research and teaching plans further at interview.

Yours sincerely,

Wentao Zhang